Politics or Print? Separating Content from Shortcuts when Classifying Two Nineteenth-Century isiXhosa Newspapers

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Abstract

We ask whether a classifier that separates *Imvo Zabantsundu*, founded by J. T. Jabavu in 1884, from the missionary *Isigidimi sama-Xosa* reads politics or print. Using the isiXhosa Intellectual Traditions corpus (Nov 1884–1888), we train tuned classifiers along a ladder that removes English, self-references, names and spelling conventions, and add non-content baselines, matching and noise injection. Raw text is separated with balanced accuracy 0.93, but spelling and transcription features alone reach 0.85. A content-word model keeps 0.81 (0.78 on noise- and spelling-matched pairs), and its features and a lexicon show Imvo using political vocabulary at about twice Isigidimi's rate. Isigidimi's political share fell after Jabavu left, but on earlier years the classifier mainly tracks time drift.

Keywords: historical isiXhosa; shortcut learning; responsible NLP

Introduction

Isigidimi sama-Xosa (Lovedale, 1870–1888) was published under missionary control. *Imvo Zabantsundu* (King William's Town, from November 1884) was founded by John Tengo Jabavu, who had edited Isigidimi from 1881 (South African History Online, 2026) and left because the missionaries were reluctant to publish explicitly political material (Odendaal, 1984). The digitised archive of Schoots et al. (2023) makes the split testable: if the break was political, the papers should differ in *what* they discuss. A source classifier looks like natural evidence, but classifiers exploit whatever separates the labels (Gururangan et al., 2018; Geirhos et al., 2020), and historical text is full of such shortcuts: collection-specific transcription error (Hill & Hengchen, 2019; van Strien et al., 2020), mastheads, adverts and house spelling.

Research question. *When a classifier separates the two papers over the years both appeared, how much of the separation is content, and how much is identity, transcription noise and orthography?* We test whether (H1) raw accuracy is inflated by shortcuts, (H2) a signal survives their removal, and (H3) that residual is political rather than religious. In a low-resource language (Joshi et al., 2020), mistaking print for politics would yield confident, false claims about historical Black writers.

Methods

Data. We use both papers from November 1884 (Imvo's first issue) to December 1888 (Isigidimi's last; its 1886 volume is lost). Pages are cut into 200-token windows (6,699 Imvo and 1,509 Isigidimi segments from 211 and 38 issues), and every cleaning level is applied to the same windows. Splits are 70/15/15 *by issue*, stratified by paper and year, for three seeds. Evaluation segments sharing over 30% of their word 6-grams with training text (reprinted adverts) are removed: 277 of 2,512.

Cleaning ladder. L0 is the full page; L1 keeps only the dataset's isiXhosa paragraphs (Imvo is 27% English, Isigidimi 1%; Schoots et al., 2023). L2 also removes self-references (titles, place of printing, staff), digits and punctuation. L3 also masks capitalised non-sentence-initial tokens (names). L4 is lower-cased, equalises orthography (⟨bh⟩→⟨b⟩, ⟨xh⟩→⟨x⟩, no elision apostrophes) and keeps the 3,485 words seen ≥ 5 times in *both* papers' training data, so one-paper spellings and OCR errors are unusable.

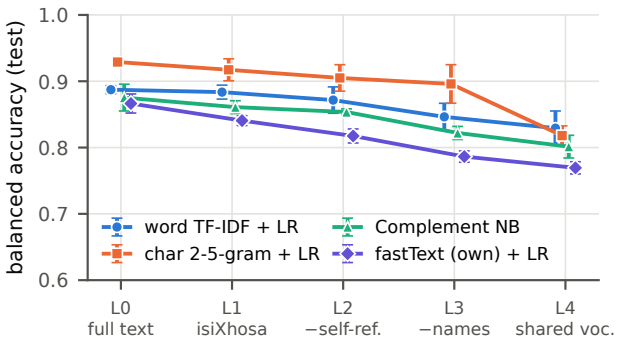


Figure 1: Test balanced accuracy (chance $= 0.5$) as shortcuts are removed; mean $\pm$ s.d. over three issue-grouped splits. The character model, which can read spelling, loses most when orthography is equalised (L3→L4).

Models. Word and character (2–5-gram) TF-IDF with class-balanced logistic regression, Complement Naive Bayes, and fastText subword embeddings (Bojanowski et al., 2017) trained on the training split and mean-pooled. Pretrained encoders (Alabi et al., 2022) could not be downloaded here. Hyperparameters are grid-searched on validation macro-F1 (scores plateau beyond the grid edge; Supp. Table 4), and each test set is used once. Isigidimi is 18% of segments, so we report macro-F1 and balanced accuracy (BA) with 95% CIs from a bootstrap over issues.

Isolating the signal. (i) *Feature families* on L4: the 100 most frequent shared words (a function-word "style" set; Mosteller & Wallace, 1964) versus the remaining "content" words. On raw text: 15 transcription-quality features (e.g. vowelless tokens, English function words) and 4 orthographic rates (⟨bh⟩, ⟨xh⟩, elision, hyphens). (ii) *Matching*: each Isigidimi test segment is paired 1:1 with its nearest Imvo segment on the 19 standardised non-content features, within a median-distance caliper (Stuart, 2010). This gives 134 pairs and cuts the mean absolute standardised difference from 0.29 to 0.08. (iii) OCR-style *noise injection* (*u*/*n*, *rn*/*m*, . . .) into Isigidimi test text; (iv) a *temporal hold-out* (test on 1888). (v) *Probe*: the content model scores unseen Isigidimi issues from 1876–80 (before Jabavu) and 1881–Oct 1884 (under Jabavu). (vi) *Lexicon*: 12 political stems (e.g. *palamente* parliament, *rulumente* government) and 12 religious stems (e.g. *Tixo* God), with false matches removed after checking concordances (Supp. S2).

Results

Shortcuts inflate accuracy (H1). On full pages character n-grams reach BA 0.93 ± 0.01 (Fig. 1). The strongest L1

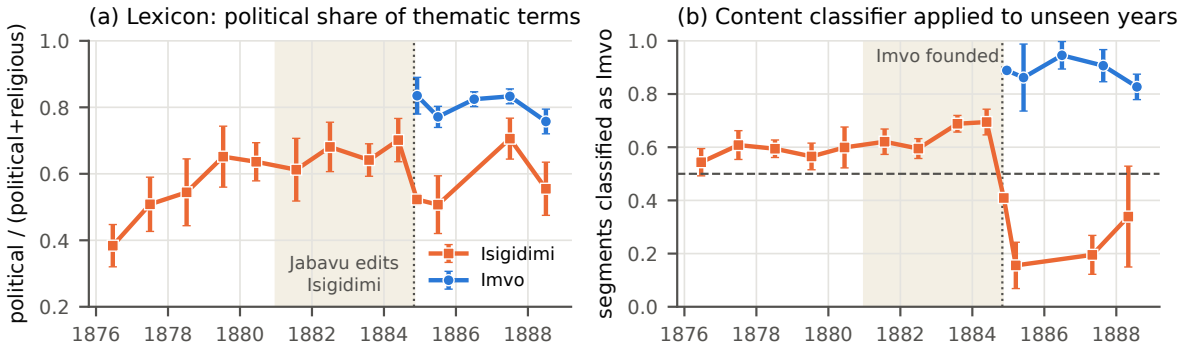


Figure 2: Isigidimi before, during (shaded) and after Jabavu's editorship, with Imvo for comparison; 1884 is split at Imvo's founding, and bars show ± 1.96 s.e. over issues. **(a)** Isigidimi's political share of thematic terms rises into the Jabavu years and falls after his departure. **(b)** Share of segments the L4 content model (trained on Nov 1884–88) labels "Imvo". Earlier years land on the Imvo side whoever the editor was, so this score tracks distance from late Isigidimi.

Table 1: Test results: macro-F1 with issue-bootstrap 95% CI (seed 13), its s.d. over 3 splits, BA, and BA_m on the 134 matched pairs. Majority-class macro-F1 ≈ 0.45.

Representation	Macro-F1 [CI]	s.d.	BA	BA_m
Full page text, char n-grams	0.926 [0.90, 0.95]	0.010	0.92	–
isiXhosa only, char n-grams	0.916 [0.88, 0.95]	0.012	0.91	0.86
isiXhosa only, words	0.883 [0.85, 0.92]	0.009	0.88	0.85
Shared vocab., words (uni+bi)	0.840 [0.79, 0.88]	0.019	0.84	0.81
Shared vocab., fastText	0.738 [0.70, 0.78]	0.002	0.78	–
Content words only	0.802 [0.76, 0.84]	–	0.81	0.78
Top-100 (function) words only	0.718 [0.68, 0.76]	–	0.75	0.71
Transcription features (GB)	0.715 [0.66, 0.77]	–	0.74	0.62
Orthography features (GB)	0.759 [0.69, 0.82]	–	0.83	0.73
All non-content features (GB)	0.831 [0.78, 0.87]	–	0.85	0.75

character features are house conventions: Imvo writes ⟨bh⟩ (8.7 tokens per 1,000 vs. 0.01) and ⟨xh⟩, while Isigidimi uses elision apostrophes three times as often. Imvo also has 2.4× more vowel-less tokens, mostly residual transcription errors. Non-content features alone reach BA 0.85 (Table 1). Disabling the reprint filter changes scores by < 0.01.

A content signal survives (H2). Each control lowers accuracy, but the L4 word model keeps BA 0.83 ± 0.03 across splits, and content words alone reach 0.81. On matched pairs the content model loses only 0.03, while the non-content models lose 0.10–0.12 (Table 1, BA_m). Matching is imperfect, so this bounds the confound. Corrupting 5% of Isigidimi tokens moves under 4% of segments to "Imvo". Tested on unseen 1888, the L4 model keeps BA 0.80. Our own fastText embeddings (about 1M training tokens) trail sparse features throughout.

The residual is political (H3). Top Imvo content features include *abamnyama* and *amawetu* ("black people", "our people"), *imantyi* (magistrate), *amatyala* (court cases), *entolongweni* (in prison) and Gordon Sprigg, Cape premier from 1886. Isigidimi's include *imbali* (history), *inkosi* (chief), *imfazwe* (war) and *Ntsikana*, the early Xhosa Christian hymn-writer. Imvo uses 14.1 political terms per 1,000 tokens [13.3, 14.8] against Isigidimi's 7.4 [6.2, 8.7] (permutation $p < 0.001$), and 3.2 vs. 5.1 religious terms. Within Isigidimi (Fig. 2a), the political share rises from 0.54 [0.49, 0.58] in 1876–80 to 0.67 [0.63, 0.70] under Jabavu, then falls to 0.59 [0.54, 0.63] after he left.

The classifier is a poor clock. On unseen Isigidimi years (Fig. 2b) the content model calls 66% of Jabavu-era segments "Imvo", but also 58% of 1876–80 segments, against 26% of held-out post-1884 ones. The Jabavu-era excess over 1876–80 is small (+0.05 mean probability, CI [0.03, 0.07]). It learned "unlike late Isigidimi", not "like Imvo".

Discussion

Print alone separates the papers (BA 0.85 without a topic word), so a 0.93 source classifier is no evidence of a political split (H1). Only after the controls does accuracy become interpretable, and it stays near 0.8 because the content differs (H2). Imvo's distinctive vocabulary is political, Isigidimi's historical and Christian (H3). This fits the account that Jabavu left to write politics freely (Odendaal, 1984): Isigidimi's political share peaked in his last years there and then fell. Its rise began in the late 1870s, however, suggesting a gradual politicisation that Isigidimi did not sustain rather than a single editor effect. We do not judge whether Jabavu was "right" to leave.

Responsible NLP. *Ablate before interpreting:* L1 feature weights would have presented ⟨bh⟩ and apostrophes as Imvo's "voice". *Noise is partly culture:* Imvo's spelling (e.g. *Xhosa*, never printed in Isigidimi) is plausibly an editorial choice; removing it suits questions of topic, not identity, and must be documented (Bender & Friedman, 2018). *Forced-choice models misread time:* a binary classifier puts every unfamiliar text into one of its two classes, so Fig. 2b does not show that 1876 Isigidimi "sounded like" Imvo. Such outputs must not become ideological labels for named authors (Blodgett et al., 2020).

Limitations. Isigidimi supplies 38 issues (about 7 per test split), so CIs are wide. The lexicons, built by a non-first-language speaker, impose a modern politics/religion split on writers for whom the two were entwined. Publication frequency and genre are confounded with source, and word division survives L4 (Imvo's separated prefixes *base*, *ase* rank highly).

Contribution Statement

This was an individual project; the author list is unchanged from the preliminary statement. Sibonelo Mlambo is responsible for all components: reading the background literature, data exploration, experiment design (hypotheses, controls, splits and metrics), implementation of the pipeline and models, analysis, and writing of this abstract, the ethics checklist and the code documentation. Compared with the preliminary statement, (i) the pretrained multilingual model was replaced by fastText embeddings trained on the corpus, because pretrained weights could not be downloaded in the compute environment; (ii) topic modelling was replaced by feature-family ablations and a validated thematic lexicon, which answered the question more directly; and (iii) non-content matching, noise injection, a temporal hold-out and an out-of-sample probe of earlier Isigidimi years were added. AI assistance was used as declared below.

Faculty of Science AI Ethics Statement

I declare that generative AI was used in this project, as permitted by the course brief. Claude (Anthropic), used through an agentic coding environment, assisted with: designing the experimental controls, writing and debugging the Python code in the submission, running the experiments, producing the figures, and drafting the text of this extended abstract, the README and the NeurIPS checklist answers. Every claim and number in the abstract was checked against the experiment outputs (`outputs/results.csv`, `outputs/stats.json`), and historical facts were checked against the cited sources. The isiXhosa glosses are limited to words whose meaning I could verify, and the thematic lexicons were checked against corpus concordances. I have read and understood all code and text in this submission and take full responsibility for it, including any errors. No AI system was used to generate data or annotations for the experiments.

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NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction claim that (i) raw-text source classification is inflated by non-content shortcuts (spelling and transcription features alone reach balanced accuracy 0.85), (ii) a content signal survives all controls (about 0.8), and (iii) this residual is political rather than religious, with Isigidimi's political share falling after Jabavu left. Each claim is tied to Table 1 or a figure. The scope is two newspapers from November 1884 to December 1888, plus earlier Isigidimi issues used only as a probe.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The Limitations paragraph and Methods note the small number of Isigidimi issues (38; wide confidence intervals), the missing 1886 Isigidimi volume, the hand-built lexicons (built by a non-first-language speaker and imposing a modern politics/religion split), the confounding of publication frequency and genre with source, word-division artefacts that survive the strictest cleaning level, imperfect matching, and the absence of pretrained encoders (which could not be downloaded in the compute environment). The Discussion also explains why the out-of-sample classifier probe cannot be read as an editor effect.

3. **Theory Assumptions and Proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [NA]

   Justification: The paper is empirical and contains no theoretical results.

4. **Experimental Result Reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: The Methods section specifies the segmentation (200-token windows), the cleaning ladder L0–L4, the issue-grouped 70/15/15 split stratified by paper and year, the near-duplicate filter, the model families and hyper-parameter grids, the matching procedure and the noise-injection procedure. All random seeds are fixed in the code.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The full code, a `requirements.txt` and a `README.txt` with the exact commands are submitted. The data are the publicly released isiXhosa Intellectual Traditions text files (Schoots et al., 2023), which were distributed for the course; the README explains where to place them.

6. **Experimental Setting/Details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

   Answer: [Yes]

Justification: The splits, grids and selection criterion (validation macro-F1, test set used once) are described in Methods; the selected hyper-parameters for every run are written to `outputs/results.csv`.

7. **Experiment Statistical Significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: All test scores carry 95% confidence intervals from a cluster bootstrap that resamples whole issues (1,000 resamples), plus the standard deviation over three independent issue splits. Lexicon differences use issue-level bootstrap intervals and a permutation test.

8. **Experiments Compute Resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: All experiments run on CPU only (2 virtual cores, 8 GB RAM, no GPU). The full pipeline (three seeds, all models including fastText) takes about 65 minutes of wall-clock time when the two commands in the README run in parallel, plus about 10 minutes for the grid-edge check; the README gives a `--fast` option that skips fastText.

9. **Code Of Ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The work uses publicly archived historical newspapers (1884–1888), involves no living human subjects, releases no generative model, and discusses the risks of attributing ideology to historical African authors from noisy text.

10. **Broader Impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: Positive: the work gives digital humanities researchers a concrete control protocol before drawing cultural conclusions from OCR-derived isiXhosa archives. Negative: source and stance classifiers trained on such data can turn digitisation artefacts and modern categories into confident but wrong claims about historical Black South African writers and their politics, and can misattribute anonymous or pseudonymous writing; the Discussion addresses this.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

    Answer: [NA]

    Justification: No new dataset or high-risk model is released; the classifiers are small linear models trained for analysis only.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

Justification: The dataset is credited to the isiXhosa Intellectual Traditions project at the University of Cape Town (Schoots et al., 2023) and was used as provided for coursework under the archive's terms; the original scans belong to the National Library of South Africa and were not used. Software (scikit-learn, gensim, NumPy, pandas, Matplotlib) is open source and cited or listed in `requirements.txt`.

13. **New Assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The only new assets are the analysis code and two small thematic lexicons (in `src/features.py`), documented in the README and in code comments with the gloss of every stem.

14. **Crowdsourcing and Research with Human Subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [NA]

    Justification: No crowdsourcing or human-subject research was performed.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [NA]

    Justification: No human participants were involved.

Supplementary Material

S1. Implementation details. Segments are 200 whitespace tokens (windows with fewer than 60 raw or 30 L4 tokens are dropped). Grids: word LR $C \in \{0.3, 1, 3, 10, 30\}$, n-grams $\{(1,1),(1,2)\}$, min. df $\{2,5\}$; char LR $C \in \{1,3,10,30\}$, $n \in \{(2,4),(2,5)\}$; Complement NB $\alpha \in \{0.01, 0.03, 0.1, 0.3, 1\}$, min. df $\{2,5\}$; fastText (skip-gram, window 5, subwords 3–5, 10 epochs, 200k hash buckets) dim $\{50,100\}$ with LR $C \in \{0.01, 0.1, 1\}$; non-content LR $C \in \{10^{-3}, \ldots, 10^{2}\}$; histogram gradient boosting learning rate $\{0.05, 0.1\}$, depth $\{2,3,5\}$. All logistic regressions are class-balanced. Selected values for every run are in `outputs/results.csv`. Software: scikit-learn (Pedregosa et al., 2011), gensim (Řehůřek & Sojka, 2010). Compute: CPU only (2 cores, 8 GB RAM); about 65 minutes wall-clock with the two README commands run in parallel.

S2. Lexicons (stems matched inside lower-cased tokens; old orthography). *Political:* palamente (parliament), voti/vota (vote), nyulo/nyulw/nyula (elect, election; excluding *-qanyul-* "cut"), rulumen (government), mteto/miteto (law), lungelo (right), rafu (tax), mantyi (magistrate), tyala (court case, charge), koloni (the Colony), bhunga (council). *Religious:* tixo (God), yesu (whole word), kristu (Christ), tandaz (pray), zibalo (scripture), vangeli (gospel), ngcwele (holy), sono (sin, word-final), mfundisi/bafundisi (minister/s), cawa (church, Sunday), nkonzo (service). Rejected after concordance checks: *kolwa* (matches *kolwandle*), *bunga* (many unrelated words), *lawul* (mostly *hlawula*, "pay").

S3. Synthetic noise. The noisier paper was identified by its rate of vowel-less plus mixed letter–digit tokens (Imvo 3.8% vs. Isigidimi 1.4% of tokens; rare-token rate was not used because it is confounded with each paper's share of the training vocabulary). Confusions were injected into Isigidimi test segments at 0–40% of tokens (Fig. 4).

Table 2: Cleaning ladder, test macro-F1 / balanced accuracy, mean ± s.d. over three issue-grouped splits (seeds 13, 21, 42).

Level	Model	Macro-F1	Bal. acc.
L0	char	0.937 ± 0.010	0.929 ± 0.007
L0	fastText	0.824 ± 0.030	0.866 ± 0.014
L0	CNB	0.812 ± 0.029	0.875 ± 0.020
L0	word	0.902 ± 0.008	0.887 ± 0.006
L1	char	0.927 ± 0.012	0.917 ± 0.017
L1	fastText	0.822 ± 0.005	0.841 ± 0.007
L1	CNB	0.857 ± 0.021	0.861 ± 0.010
L1	word	0.891 ± 0.009	0.884 ± 0.011
L2	char	0.916 ± 0.017	0.905 ± 0.020
L2	fastText	0.796 ± 0.009	0.817 ± 0.011
L2	CNB	0.852 ± 0.016	0.854 ± 0.005
L2	word	0.886 ± 0.013	0.872 ± 0.020
L3	char	0.906 ± 0.025	0.896 ± 0.029
L3	fastText	0.764 ± 0.013	0.786 ± 0.009
L3	CNB	0.819 ± 0.013	0.822 ± 0.010
L3	word	0.859 ± 0.016	0.846 ± 0.020
L4	char	0.816 ± 0.007	0.818 ± 0.015
L4	fastText	0.740 ± 0.002	0.769 ± 0.009
L4	CNB	0.767 ± 0.009	0.801 ± 0.017
L4	word	0.839 ± 0.019	0.829 ± 0.026

Table 3: Robustness (seed 13). *No dedup*: near-duplicate filter switched off. *Temporal*: train/validate on Nov 1884–1887, test on 1888. Reference: the main random split.

Run	Level	Model	Macro-F1	Ref. macro-F1
No dedup	L0	word	0.902	0.899
No dedup	L0	char	0.928	0.926
No dedup	L1	word	0.890	0.883
No dedup	L1	char	0.920	0.916
Temporal	L1	word	0.864	0.883
Temporal	L1	char	0.896	0.916
Temporal	L4	word	0.787	0.840
Temporal	L4	char	0.745	0.811

Table 4: Grid-edge check (validation macro-F1, seed 13). Several selected C values sat at the top of the grid; larger values do not improve validation scores.

Level	Model	C=10	30	100	300
L1	char	0.944	0.948	0.946	0.946
L1	word	0.885	0.888	0.890	0.886
L4	char	0.824	0.813	0.798	0.795
L4	word	0.836	0.824	0.817	0.815

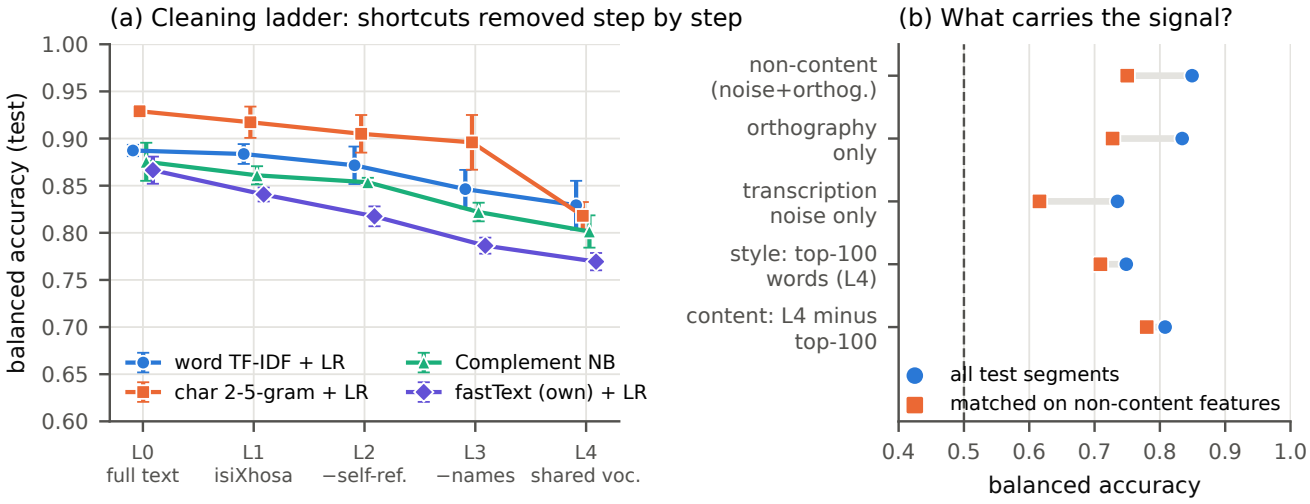


Figure 3: Extended version of Fig. 1. **(b)** Single feature families (seed 13) on all test segments (blue) and on the 134 pairs matched for the 19 non-content features (orange): matching costs the content model 0.03 but the non-content models 0.10–0.12.

Table 5: Editorial eras. Lexicon rates per 1,000 tokens (issue-level means, bootstrap 95% CI over issues; all issues). Probe: mean P(Imvo) from the L4 content model and share of segments classified Imvo (post-1884 rows: held-out test issues only).

Group	Issues (lex.)	Political/1k	Religious/1k	Political share	Mean P(Imvo)	% segs. Imvo
Isigidimi 1876-80 (pre-Jabavu)	51	8.93 [8.12, 9.80]	8.23 [7.08, 9.40]	0.54 [0.49, 0.58]	0.55	58
Isigidimi 1881-Oct 1884 (Jabavu)	56	10.04 [9.00, 11.12]	4.76 [4.22, 5.34]	0.67 [0.63, 0.70]	0.60	66
Isigidimi Nov 1884-88 (post-Jabavu)	38	7.38 [6.22, 8.73]	5.14 [4.32, 5.98]	0.59 [0.54, 0.63]	0.34	26
Imvo Nov 1884-88	211	14.06 [13.32, 14.79]	3.23 [3.01, 3.47]	0.80 [0.78, 0.81]	0.78	88

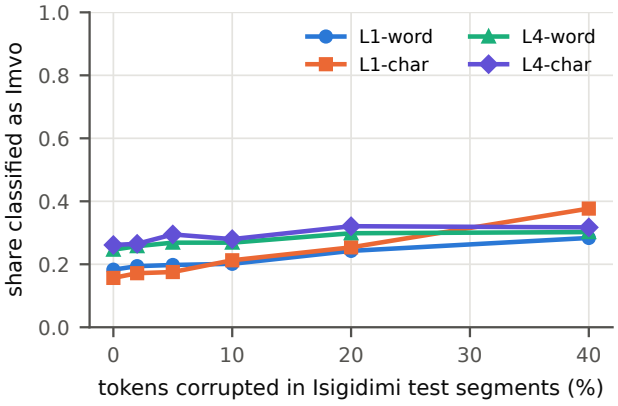


Figure 4: Share of Isigidimi test segments classified as Imvo as synthetic OCR confusions are injected. Realistic rates (2–5%) barely move predictions; only implausibly heavy corruption shifts the raw character model.

Table 6: Twenty largest-weight features of the L4 content model, in order (seed 13). Glosses for selected items are given in the Results.

Favour Imvo	Favour Isigidimi
njengokuba, abamnyama, mayelana, base, ase, uku, sprigg, egqitileyo, kukuba, ba, zetu, epepeni, suka, omnyama, ixabiso, oka, okokuba, lokuba, amawetu, kwelilizwe	*imbali, ed, ngoko, ngosuku, malunga, uzele, suke, ezinje, londawo, inani, um, kanjalo, mbali, evekini, umlambo, ngolwe, ingu, ngxoxo, kwase, ndawo*

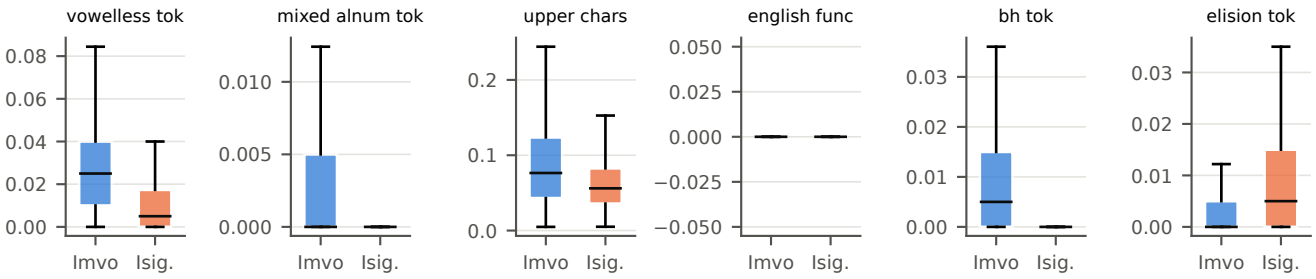


Figure 5: Selected non-content features per segment (training split; boxes: interquartile range, whiskers: 1.5 IQR). Imvo has more vowel-less and mixed letter–digit tokens, uppercase characters and English function words; the ⟨bh⟩ digraph is almost exclusive to Imvo and apostrophe elision is more frequent in Isigidimi.

APPENDIX B:

Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools ☐

I **did** make use of generative AI tools for the following (tick all that apply):

1. Idea Generation (research problem/design, hypothesis) ☑
2. Sourcing Related Work (summarising, identifying sources) ☑
3. Methods and Experiment Design (experiment setup, model tuning) ☑
4. Data Analysis (presentation, coding, interpretation) ☑
5. Theoretical Development (theorem proving, conceptual analysis) ☐
6. Code Development (generating algorithms, writing scripts) ☑
7. Presentation (rendering graphics, formatting) ☑
8. Editing (grammar, readability) ☑
9. Writing (text generation, document structuring) ☑
10. Citation Formatting (structuring, organising) ☑

If other uses were involved, please specify below:

Generative AI tool used (list all)	**Used for?**
Claude (Anthropic)	Items 1-4, 6-10 above; see AI Ethics Statement

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

Student number: 536187

Candidate signature: Sibonelo_Mlambo

Date: 2026-10-29